

JAKUB BIAŁAS

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PROFILE

Graduate in Data Science and Bioinformatics, with experience in data analysis, statistical modeling, survival analysis, and machine learning. My project experience includes oncology, data augmentation, predictive model evaluation, data cleaning, and data visualization. I work in Python (pandas, numpy, scikit-learn) and R, conducting data analysis and statistical experiments.

Interested in roles such as Data Analyst, Junior Data Scientist, or Bioinformatics Analyst.

SKILLS

- **Programming languages:** Python (pandas, numpy, scikit-learn – data analysis, machine learning, feature engineering), R (statistical analysis, survival analysis)
- **Databases:** SQL (PostgreSQL, MySQL – joins, data aggregation, data analysis), basics of ETL/ELT pipelines (data extraction, transformation, loading)
- **Data analysis & visualization:** Power BI, Tableau (dashboard creation, KPI reporting, data visualization), Excel (pivot tables, data analysis, reporting)
- **Tools & environments:** Git (version control), Linux (basic CLI operations), Jupyter Notebook, Microsoft 365 (Excel, PowerPoint, Word)
- **Languages:** Polish (native), English (B2)

EDUCATION

- **02/2024 – 09/2025**

Silesian University of Technology, Faculty of Automatic Control, Electronics and Computer Science
Master's degree, Automation, Electronics and Computer Science, specialization: Data Science -

Scope: optimization methods, statistics, artificial intelligence, machine learning, data analysis, data visualization.

- **10/2020 – 02/2024**

Silesian University of Technology, Faculty of Automatic Control, Electronics and Computer Science
Bachelor's degree in Engineering, Biotechnology, specialization: Bioinformatics -

Scope: bioinformatics, biochemistry, cell biology, genetics, biochemical processes, laboratory and analytical methods.

- **09/2017 – 06/2020**

IV General Secondary School named after Maria Skłodowska-Curie in Chorzów – Graduated with distinction. Advanced subjects: mathematics, chemistry, biology.

ACHIEVEMENTS, PROJECTS

- **09/2025 – Master's thesis entitled “Augmentation of medical data on response to cancer treatment in small clinical trials”**

- Compared 8 survival data augmentation methods, including bootstrap, SMOTE, Cox model-based approaches, and an original Cox-SMOTE method.
- Developed an original Cox-SMOTE method for generating synthetic survival data for small clinical trials.
- Built an experimental framework to evaluate methods using TES, $\Delta RMST$, Spearman and Pearson correlations, variance analysis, and regression parameters.
- Conducted 10 repetitions for each augmentation scenario to assess the stability and reproducibility of results.
- Demonstrated that methods accounting for the structure of survival data, particularly Cox-SMOTE, improve the relationship between TES and $\Delta RMST$.

- **01/2024 – Engineering project entitled “Modification and extension of algorithms for estimating spatial heterogeneity of lymphocyte maps in cancer prognosis”**

- Developed a GLCM-based histopathological image analysis algorithm for cancer prognosis.
- Designed an image preprocessing pipeline including filtering, morphological operations, background removal, image cropping, and grayscale level adjustment.
- Evaluated the impact of parameter modifications on prediction quality using the Cox model, Harrell's C-index, and p-value.
- Identified configurations that improved model effectiveness; in the best case, increased the C-index from 0.72 to 0.91.

- **07/2023 – 08/2023 – Student internship at the Klimzowiec wastewater treatment plant**

- Monitored wastewater treatment plant operating parameters in the SCADA system.
- Supported laboratory processes related to sample handling.
- Helped prepare assumptions for the renovation of a technological tank.

- **06/2022 – Cambridge First Certificate (FCE)**

Jakub Białas